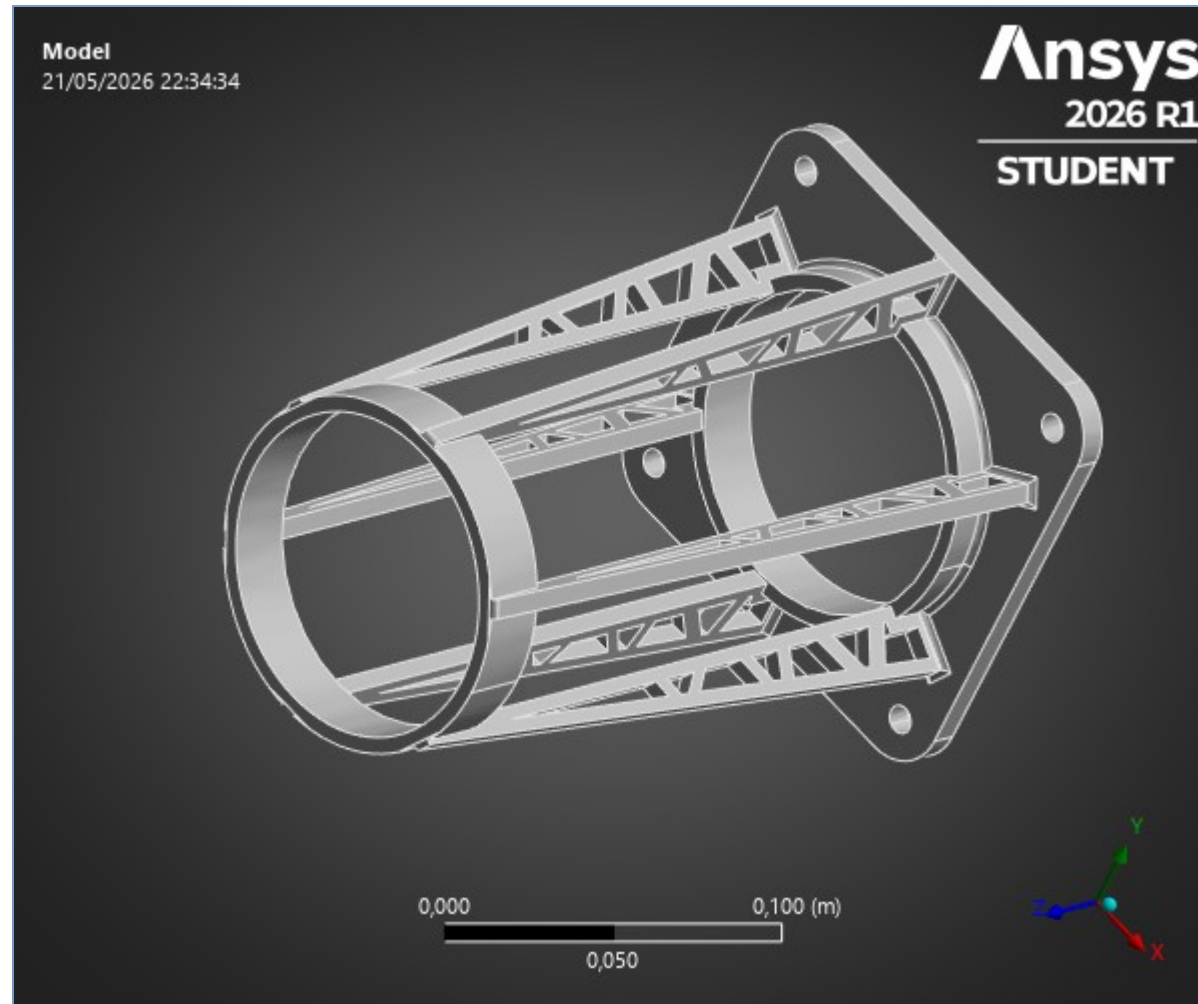




Project

First Saved	Thursday, May 21, 2026
Last Saved	Thursday, May 21, 2026
Product Version	2026 R1
Save Project Before Solution	No
Save Project After Solution	No



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Units

TABLE 1

Unit System	Metric (m, kg, N, s, V, A) Degrees rad/s Celsius
Angle	Degrees
Rotational Velocity	rad/s
Temperature	Celsius

Model (A4)

TABLE 2	
Model (A4) > Geometry Imports	
Object Name	<i>Geometry Imports</i>

State	Solved
-------	--------

TABLE 3
Model (A4) > Geometry Imports > Geometry Import (A3)

Object Name	<i>Geometry Import (A3)</i>
State	Solved
Definition	
Source	C:\Users\Mateus\OneDrive\Documentos\UnB\4 sem\Projetos\Engine mount\structure support test 1.stp
Type	Step
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No
Use Instances	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
Enclosure and Symmetry Processing	Yes

Geometry

TABLE 4
Model (A4) > Geometry

--	--

Object Name	Geometry
State	Fully Defined
Definition	
Source	C:\Users\Mateus\OneDrive\Documentos\UnB\4 sem\Projetos\Engine mount\structure support test 1.stp
Type	Step
Length Unit	Millimeters
Element Control	Program Controlled
Display Style	Body Color
Bounding Box	
Length X	0,145 m
Length Y	0,145 m
Length Z	0,224 m
Properties	
Volume	2,2549e-004 m³
Mass	0,62461 kg
Scale Factor Value	1,
Statistics	
Bodies	1
Active Bodies	1
Nodes	12285
Elements	5195
Mesh Metric	None
Update Options	
Assign Default Material	No
Basic Geometry Options	
Solid Bodies	Yes
Surface Bodies	Yes
Line Bodies	No
Parameters	Independent
Parameter Key	ANS;DS
Attributes	No
Named Selections	No
Material Properties	No
Advanced Geometry Options	
Use Associativity	Yes
Coordinate Systems	No
Reader Mode Saves Updated File	No

	Yes
Smart CAD Update	Yes
Compare Parts On Update	No
Analysis Type	3-D
Mixed Import Resolution	None
Import Facet Quality	Source
Clean Bodies On Import	No
Stitch Surfaces On Import	None
Decompose Disjoint Geometry	Yes
ID_GeometryPrefProcessPhysicsDefinition	No
Enclosure and Symmetry Processing	Yes

TABLE 5
Model (A4) > Geometry > Parts

Object Name	<i>Part2 PartBody</i>
State	Meshed
Graphics Properties	
Visible	Yes
Transparency	1
Definition	
Suppressed	No
Stiffness Behavior	Flexible
Coordinate System	Default Coordinate System
Reference Temperature	By Environment
Treatment	None
Material	
Assignment	Aluminum Alloy
Nonlinear Effects	Yes
Thermal Strain Effects	Yes
Bounding Box	
Length X	0,145 m
Length Y	0,145 m
Length Z	0,224 m
Properties	
Volume	2,2549e-004 m ³
Mass	0,62461 kg
Centroid X	5,9677e-006 m

Centroid Y	5,9677e-006 m
Centroid Z	6,3831e-002 m
Moment of Inertia Ip1	5,2225e-003 kg·m ²
Moment of Inertia Ip2	5,2228e-003 kg·m ²
Moment of Inertia Ip3	2,377e-003 kg·m ²
Statistics	
Nodes	12285
Elements	5195
Mesh Metric	None

TABLE 6
Model (A4) > Materials

Object Name	<i>Materials</i>
State	Fully Defined
Statistics	
Materials	2
Material Assignments	0

Coordinate Systems

TABLE 7
Model (A4) > Coordinate Systems > Coordinate System

Object Name	<i>Global Coordinate System</i>
State	Fully Defined
Definition	
Type	Cartesian
Coordinate System ID	0,
Origin	
Origin X	0, m
Origin Y	0, m
Origin Z	0, m
Directional Vectors	
X Axis Data	[1, 0, 0,]
Y Axis Data	[0, 1, 0,]
Z Axis Data	[0, 0, 1,]
Transfer Properties	
Source	

Read Only	No
-----------	----

Mesh

TABLE 8
Model (A4) > Mesh

Object Name	<i>Mesh</i>
State	Solved
Display	
Display Style	Use Geometry Setting
Defaults	
Physics Preference	Mechanical
Element Order	Program Controlled
Element Size	Default
Sizing	
Mesh Sizing	Adaptive
Resolution	Default (2)
Mesh Defeaturing	Yes
Defeature Size	Default
Transition	Fast
Span Angle Center	Coarse
Initial Size Seed	Assembly
Bounding Box Diagonal	0,30369 m
Average Surface Area	4,0604e-004 m ²
Minimum Edge Length	1,6072e-004 m
Quality	
Check Mesh Quality	Yes, Errors
Error Limits	Aggressive Mechanical
Target Element Quality	Default (0,1)
Smoothing	Medium
Mesh Metric	None
Inflation	
Use Automatic Inflation	None
Inflation Option	Smooth Transition
Transition Ratio	0,272
Maximum Layers	5
Growth Rate	1,2

Inflation Algorithm	Pre
Inflation Element Type	Wedges
View Advanced Options	No
Advanced	
Number of CPUs for Parallel Part Meshing	Program Controlled
Straight Sided Elements	No
Rigid Body Behavior	Dimensionally Reduced
Triangle Surface Mesher	Program Controlled
Topology Checking	Yes
Pinch Tolerance	Please Define
Generate Pinch on Refresh	No
Auto-Map Fillets	No
Automatic Methods	
Sheet Body Method	Prime Quad Dominant
Sweepable Body Method	Sweep
Statistics	
Nodes	12285
Elements	5195
Show Detailed Statistics	No

Static Structural (A5)

TABLE 9
Model (A4) > Analysis

Object Name	<i>Static Structural (A5)</i>
State	Solved
Definition	
Physics Type	Structural
Analysis Type	Static Structural
Solver Target	Mechanical APDL
Options	
Environment Temperature	22, °C
Generate Input Only	No

TABLE 10
Model (A4) > Static Structural (A5) > Analysis Settings

Object Name	<i>Analysis Settings</i>
-------------	--------------------------

State	Fully Defined
Step Controls	
Number Of Steps	1,
Current Step Number	1,
Step End Time	1, s
Auto Time Stepping	Program Controlled
Maximum Iterations Per Substep	Program Controlled
Solver Controls	
Solver Type	Program Controlled
Weak Springs	Off
Solver Pivot Checking	Program Controlled
Large Deflection	Off
Inertia Relief	Off
Quasi-Static Solution	Off
Rotordynamics Controls	
Coriolis Effect	Off
Restart Controls	
Generate Restart Points	Program Controlled
Retain Files After Full Solve	No
Combine Restart Files	Program Controlled
Nonlinear Controls	
Newton-Raphson Option	Program Controlled
Force Convergence	Program Controlled
Moment Convergence	Program Controlled
Displacement Convergence	Program Controlled
Rotation Convergence	Program Controlled
Line Search	Program Controlled
Stabilization	Program Controlled
Advanced	
Inverse Option	No
Contact Split (DMP)	Program Controlled
Output Controls	
Output Selection	None
Stress	Yes
Back Stress	No
Strain	Yes
Contact Data	Yes

	No
Nodal Forces	No
Volume and Energy	Yes
Euler Angles	Yes
General Miscellaneous	No
Contact Miscellaneous	No
Store Results At	All Time Points
Result File Options	
Result File Compression	Program Controlled
Analysis Data Management	
Solver Files Directory	C:\Users\Mateus\Engine-Mount-Optimization\Ansys\test\test project_files\dp0\SYS\MECH\
Future Analysis	None
Scratch Solver Files Directory	
Save MAPDL db	No
Contact Summary	Program Controlled
Delete Unneeded Files	Yes
Nonlinear Solution	No
Solver Units	Active System
Solver Unit System	mks

TABLE 11
Model (A4) > Static Structural (A5) > Loads

Object Name	Fixed Support	Force
State	Fully Defined	
Scope		
Scoping Method	Geometry Selection	
Geometry	8 Faces	1 Face
Definition		
Type	Fixed Support	Force
Suppressed	No	
Define by		Components
Applied By		Surface Effect
Coordinate System		Global Coordinate System
X Component		0, N (ramped)
Y Component		0, N (ramped)
Z Component		6750, N (ramped)

Solution (A6)

TABLE 12
Model (A4) > Static Structural (A5) > Solution

Object Name	<i>Solution (A6)</i>
State	Solved
Adaptive Mesh Refinement	
Max Refinement Loops	1,
Refinement Depth	2,
Information	
Status	Done
MAPDL Elapsed Time	16,552 s
MAPDL Memory Used	232, MB
MAPDL Result File Size	3,6875 MB
Post Processing	
Beam Section Results	No
On Demand Stress/Strain	No

TABLE 13
Model (A4) > Static Structural (A5) > Solution (A6) > Solution Information

Object Name	<i>Solution Information</i>
State	Solved
Solution Information	
Solution Output	Solver Output
Newton-Raphson Residuals	0
Identify Element Violations	0
Update Interval	2,5 s
Display Points	All
FE Connection Visibility	
Activate Visibility	Yes
Display	All FE Connectors
Draw Connections Attached To	All Nodes
Line Color	Connection Type
Visible on Results	No
Line Thickness	Single
Display Type	Lines

TABLE 14
Model (A4) > Static Structural (A5) > Solution (A6) > Results

Object Name	Total Deformation	Equivalent Stress
State	Solved	
Scope		
Scoping Method	Geometry Selection	
Geometry	All Bodies	
Definition		
Type	Total Deformation	Equivalent (von-Mises) Stress
By	Time	
Display Time	Last	
Separate Data by Entity	No	
Calculate Time History	Yes	
Identifier		
Suppressed	No	
Results		
Minimum	0, m	1,8933e+005 Pa
Maximum	1,196e-004 m	6,8448e+007 Pa
Average	6,73e-005 m	1,6342e+007 Pa
Minimum Occurs On	Part2 PartBody	
Maximum Occurs On	Part2 PartBody	
Information		
Time	1, s	
Load Step	1	
Substep	1	
Iteration Number	1	
Integration Point Results		
Display Option		Averaged
Average Across Bodies		No

FIGURE 1
Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Model (A4) > Static Structural (A5) > Solution (A6) > Total Deformation

Time [s]	Minimum [m]	Maximum [m]	Average [m]
1,	0,	1,196e-004	6,73e-005

Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

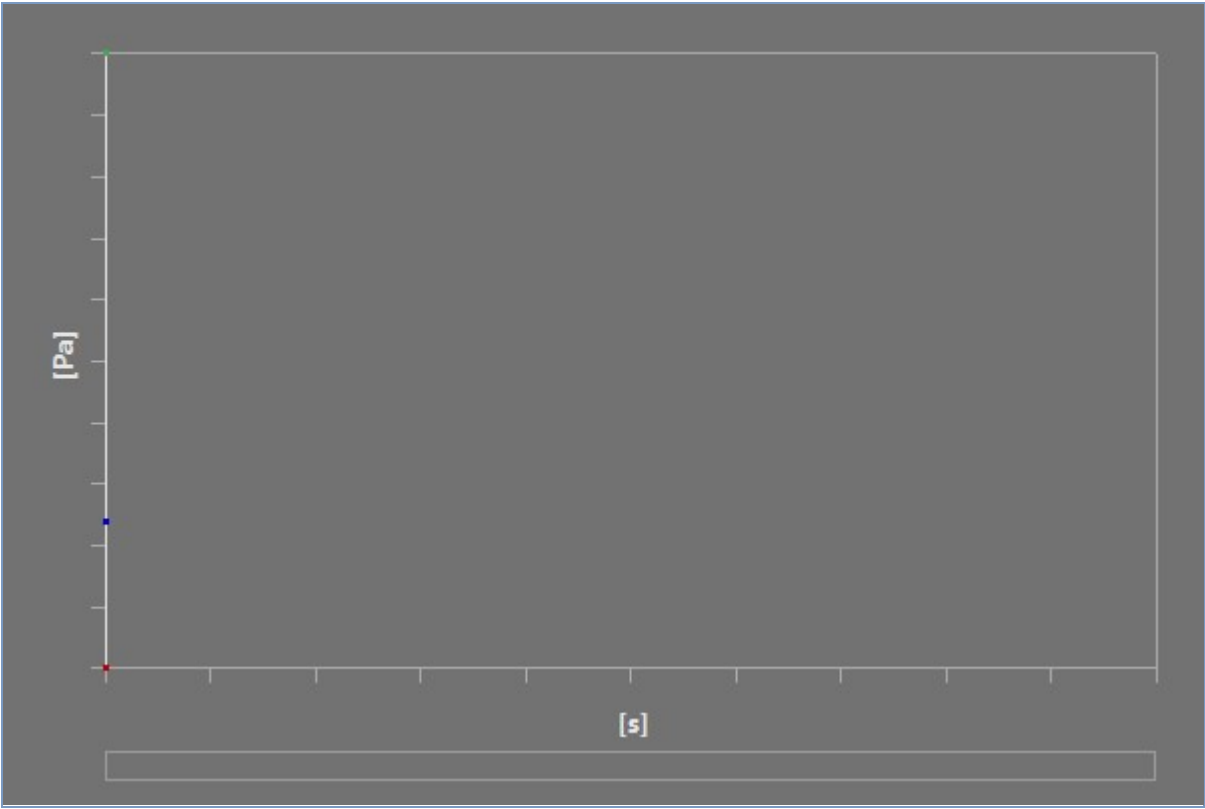


TABLE 16
Model (A4) > Static Structural (A5) > Solution (A6) > Equivalent Stress

Time [s]	Minimum [Pa]	Maximum [Pa]	Average [Pa]
1,	1,8933e+005	6,8448e+007	1,6342e+007

Material Data

Aluminum Alloy

TABLE 17
Aluminum Alloy > Constants

Density	2770, kg m^-3
---------	---------------

Coefficient of Thermal Expansion	2,3e-005 C ⁻¹
Specific Heat	875, J kg ⁻¹ C ⁻¹

TABLE 18
Aluminum Alloy > Color

Red	Green	Blue
138,	104,	46,

TABLE 19
Aluminum Alloy > Compressive Ultimate Strength

Compressive Ultimate Strength Pa
0,

TABLE 20
Aluminum Alloy > Compressive Yield Strength

Compressive Yield Strength Pa
2,8e+008

TABLE 21
Aluminum Alloy > Tensile Yield Strength

Tensile Yield Strength Pa
2,8e+008

TABLE 22
Aluminum Alloy > Tensile Ultimate Strength

Tensile Ultimate Strength Pa
3,1e+008

TABLE 23
Aluminum Alloy > Isotropic Secant Coefficient of Thermal Expansion

Zero-Thermal-Strain Reference Temperature C
22,

TABLE 24
Aluminum Alloy > Isotropic Thermal Conductivity

Thermal Conductivity W m ⁻¹ C ⁻¹	Temperature C
114,	-100,
144,	0,

165,	100,
175,	200,

TABLE 25
Aluminum Alloy > S-N Curve

Alternating Stress Pa	Cycles	R-Ratio
2,758e+008	1700,	-1,
2,413e+008	5000,	-1,
2,068e+008	34000	-1,
1,724e+008	1,4e+005	-1,
1,379e+008	8,e+005	-1,
1,172e+008	2,4e+006	-1,
8,963e+007	5,5e+007	-1,
8,274e+007	1,e+008	-1,
1,706e+008	50000	-0,5
1,396e+008	3,5e+005	-0,5
1,086e+008	3,7e+006	-0,5
8,791e+007	1,4e+007	-0,5
7,757e+007	5,e+007	-0,5
7,239e+007	1,e+008	-0,5
1,448e+008	50000	0,
1,207e+008	1,9e+005	0,
1,034e+008	1,3e+006	0,
9,308e+007	4,4e+006	0,
8,618e+007	1,2e+007	0,
7,239e+007	1,e+008	0,
7,412e+007	3,e+005	0,5
7,067e+007	1,5e+006	0,5
6,636e+007	1,2e+007	0,5
6,205e+007	1,e+008	0,5

TABLE 26
Aluminum Alloy > Isotropic Resistivity

Resistivity kg m ³ A ⁻² s ⁻³	Temperature C
2,43e-008	0,
2,67e-008	20,
3,63e-008	100,

TABLE 27
Aluminum Alloy > Isotropic Elasticity

Young's Modulus Pa	Poisson's Ratio	Bulk Modulus Pa	Shear Modulus Pa	Temperature C
7,1e+010	0,33	6,9608e+010	2,6692e+010	

TABLE 28
Aluminum Alloy > Isotropic Relative Permeability

Relative Permeability
1,